Test Case Specification (v1.1)

Field	Detail
Project	Vehicle Parking System (VPS)
Version	1.1 — updated after Sprints 1–2 (Part 3)
Supersedes	Test cases in Software Test Plan v1.0 §13 (16-09-2026)
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This specification lists every test case, the SRS requirement it verifies, the steps and expected result, what changed during the sprints, and the automated tests that implement it. The status column is taken from the latest run. Total: **38 test cases** (28 functional, 10 non-functional / security).

1. Summary

Test case	Title	Requirement(s)	Type	Priority	Status
TC-ENT-01	ANPR plate capture	VPS-F-001	Functional / Performance	High	PASS
TC-ENT-02	QR ticket issue	VPS-F-002, VPS-SR-004	Functional	High	PASS
TC-ENT-03	Entry barrier raise	VPS-F-003	Functional / Performance	High	PASS
TC-ENT-04	Ground-loop safety halt	VPS-F-004	Functional / Safety	High	PASS
TC-ENT-05	Anti-passback	VPS-F-002	Functional	Medium	PASS
TC-ENT-06	Invalid / unreadable plate	VPS-F-001	Negative	High	PASS
TC-SLOT-01	Nearest vacant slot by category	VPS-F-005	Functional	High	PASS
TC-SLOT-02	Sensor → Occupied + red LED	VPS-F-006	Functional / Performance	High	PASS
TC-SLOT-03	VMS boards match database	VPS-F-007	Functional	Medium	PASS
TC-SLOT-04	Lot full / concurrent allocation	VPS-F-005	Negative / Concurrency	High	PASS
TC-BILL-01	Tiered fare computation	VPS-F-008	Functional	High	PASS
TC-BILL-02	15-minute grace period	VPS-F-009	Boundary	Medium	PASS
TC-BILL-03	Surge and coupons	VPS-F-010	Functional	Medium	PASS
TC-PAY-01	Exit scan retrieves invoice	VPS-F-011	Functional / Performance	High	PASS
TC-PAY-02	Cashless payment	VPS-F-012	Functional / Integration	High	PASS
TC-PAY-03	Exit barrier on PAID only	VPS-F-013	Functional / Security	High	PASS

Test case	Title	Requirement(s)	Type	Priority	Status
TC-PAY-04	GST tax receipt	VPS-F-014	Functional	Medium	PASS
TC-PAY-05	Payment failure & webhook security	VPS-F-012, VPS-F-013	Negative / Security	High	PASS
TC-RES-01	48-hour advance reservation	VPS-F-015	Functional / Boundary	High	PASS
TC-RES-02	No-show auto-release	VPS-F-016	Functional / Boundary	Medium	PASS
TC-RES-03	Reservation check-in at entry	VPS-F-015	Functional / Negative	Medium	PASS
TC-ADM-01	Admin configuration without restart	VPS-F-017	Functional	High	PASS
TC-ADM-02	Report export <= 5 s	VPS-F-018	Functional / Performance	Medium	PASS
TC-Perf-01	Ingress latency p90	VPS-NF-001	Performance	High	PASS
TC-Rel-01	Availability proxy (soak)	VPS-NF-002	Reliability	High	PASS
TC-Scale-01	Scale: 10k slots / 50 lanes	VPS-NF-003	Performance / Load	High	PASS
TC-SEC-06	PCI-DSS: no plaintext card data	VPS-NF-004	Security	High	PASS
TC-Fault-01	Edge offline resilience	VPS-NF-005	Resilience	High	PASS
TC-SEC-01	TLS 1.3 enforced	VPS-SR-001	Security	High	PASS
TC-SEC-02	Password hashing & PII encryption	VPS-SR-002	Security	High	PASS
TC-SEC-03	RBAC + MFA	VPS-SR-003	Security	High	PASS
TC-SEC-04	Counterfeit QR rejected	VPS-SR-004	Security	High	PASS
TC-SEC-05	Immutable audit log	VPS-SR-005	Security	High	PASS
TC-SYS-01	Walk-in car, UPI payment	VPS-F-001, VPS-F-005, VPS-F-008, VPS-F-012, VPS-F-013	System (E2E)	High	PASS
TC-SYS-02	Reserved EV, short visit	VPS-F-015, VPS-F-009	System (E2E)	Medium	PASS
TC-SYS-03	Peak-hour card payment with coupon	VPS-F-010, VPS-F-012	System (E2E)	Medium	PASS
TC-SYS-04	Unreadable plate, cash exit	VPS-F-001, VPS-F-012	System (E2E)	Medium	PASS
TC-SYS-05	Admin day	VPS-F-017, VPS-F-018	System (E2E)	Medium	PASS

2. Test case details

TC-ENT-01 — ANPR plate capture

Field	Detail
Requirement(s)	VPS-F-001 (Capture vehicle plate via ANPR within 1.5 s (>= 95 % OCR accuracy))

Field	Detail
Type / priority	Functional / Performance / High
Preconditions	Entry lane ENTRY-1 online; camera online
Steps	1. Vehicle arrives with plate 'ka-01 ab 1234'. 2. Camera reads plate at confidence 0.97. 3. Observe the read.
Expected result	Plate normalised to KA01AB1234 and validated against Indian formats (incl. BH series); capture <= 1500 ms.
Sprint update	Added normalisation of spaces/hyphens and BH-series plates.

Automated by	`unit/test_hardware.py::TestCamera::test_capture_ok`
	`unit/test_hardware.py::TestCamera::test_confidence_range`
	`unit/test_hardware.py::TestCamera::test_low_confidence`
	`unit/test_hardware.py::TestCamera::test_offline`
	`unit/test_hardware.py::test_plate_normalisation`
	`unit/test_ingress.py::test_plate_capture_normalised_and_fast`

Latest result	**PASS** (9/9 automated tests passed)

TC-ENT-02 — QR ticket issue

Field	Detail
Requirement(s)	VPS-F-002 (Issue entry ticket with unique QR, signed session ID and timestamp within 2 s), VPS-SR-004 (HMAC-SHA256 signed QR tickets; tampered tickets rejected)
Type / priority	Functional / High
Preconditions	Valid plate read; vacant slot of the category
Steps	1. POST /api/v1/ingress/ticket {plate_number, vehicle_category, timestamp}. 2. Verify QR signature. 3. Inspect DB row.
Expected result	201 with ticket_id (unique UUID), qr_payload (HMAC-SHA256 signed: ticket id, slot, timestamp), allocated_slot; issued <= 2000 ms; plate stored AES-encrypted; ticket printed.
Sprint update	Response shape aligned with SAD 4.3; printer-fault fallback to digital ticket added.

Automated by	`unit/test_ingress.py::test_api_matches_sad_contract`
	`unit/test_ingress.py::test_ticket_has_signed_qr_and_encrypted_plate`
	`unit/test_ingress.py::test_ticket_ids_unique`

Latest result	**PASS** (3/3 automated tests passed)

TC-ENT-03 — Entry barrier raise

Field	Detail
Requirement(s)	VPS-F-003 (Raise entry barrier within 500 ms on valid ticket)
Type / priority	Functional / Performance / High
Preconditions	Ticket issued
Steps	1. Issue ticket. 2. Read barrier state and relay latency.
Expected result	Barrier UP; relay command <= 500 ms; total ingress <= 3000 ms.
Sprint update	RFID path deferred (DEV-07).

Automated by	`unit/test_hardware.py::TestBarrier::test_halt_on_loop`
	`unit/test_hardware.py::TestBarrier::test_lane_auto_lower_after_vehicle_clears`
	`unit/test_hardware.py::TestBarrier::test_raise_and_lower`
	`unit/test_hardware.py::TestBarrier::test_relay_fault`
	`unit/test_ingress.py::test_barrier_raised_within_500ms`

Latest result	**PASS** (5/5 automated tests passed)

TC-ENT-04 — Ground-loop safety halt

Field	Detail
Requirement(s)	VPS-F-004 (Prevent barrier descent while vehicle on ground loop)
Type / priority	Functional / Safety / High
Preconditions	Barrier UP after a ticket
Steps	1. Loop reports vehicle present. 2. Command barrier down. 3. Loop clears.
Expected result	Step 2: descent halted (reason VEHICLE_ON_LOOP), barrier stays UP. Step 3: barrier lowers automatically.
Sprint update	Auto-lower on loop clear added.

Automated by	`unit/test_hardware.py::TestBarrier::test_halt_on_loop`
	`unit/test_hardware.py::TestBarrier::test_lane_auto_lower_after_vehicle_clears`
	`unit/test_hardware.py::TestBarrier::test_raise_and_lower`
	`unit/test_hardware.py::TestBarrier::test_relay_fault`
	`unit/test_ingress.py::test_api_loop_safety`

Latest result	**PASS** (5/5 automated tests passed)

TC-ENT-05 — Anti-passback

Field	Detail
Requirement(s)	VPS-F-002 (Issue entry ticket with unique QR, signed session ID and timestamp within 2 s)
Type / priority	Functional / Medium
Preconditions	Vehicle KA01AB1234 already inside
Steps	1. Same plate arrives at ENTRY-2.
Expected result	409 VEHICLE_ALREADY_INSIDE; no second ticket.
Sprint update	New case (Sprint 1).
Automated by	`unit/test_ingress.py::test_anti_passback`
Latest result	**PASS** (1/1 automated tests passed)

TC-ENT-06 — Invalid / unreadable plate

Field	Detail
Requirement(s)	VPS-F-001 (Capture vehicle plate via ANPR within 1.5 s (>= 95 % OCR accuracy))
Type / priority	Negative / High
Preconditions	Entry lane online
Steps	1. Submit malformed plates ('', 'HELLO', 'KA01AB12345'). 2. Submit valid plate with confidence 0.5. 3. Attendant enters plate manually.
Expected result	1: 400 INVALID_PLATE. 2: 422 PLATE_UNREADABLE (routed to attendant). 3: ticket with entry_mode MANUAL, audited.
Sprint update	New case (Sprint 1): attendant manual-entry path.
Automated by	`unit/test_ingress.py::test_invalid_plate` `unit/test_ingress.py::test_low_confidence_routes_to_attendant`
Latest result	**PASS** (5/5 automated tests passed)

TC-SLOT-01 — Nearest vacant slot by category

Field	Detail
Requirement(s)	VPS-F-005 (Designate nearest vacant slot by vehicle category)
Type / priority	Functional / High
Preconditions	Default layout seeded (100 bays, 3 levels)

Field	Detail
Steps	1. Allocate one slot for each of 2W, 4W, EV, HANDICAP. 2. Allocate twice more; release first; allocate again. 3. Exhaust level-1 EV bays.
Expected result	Slot category matches; chosen slot has the lowest (level, distance); released slot is reused first; allocation spills to level 2.
Sprint update	Unchanged.

Automated by	`unit/test_allocation.py::test_allocation_moves_to_next_nearest`
	`unit/test_allocation.py::test_nearest_slot_matches_category`
	`unit/test_allocation.py::test_spills_to_next_level`

Latest result	**PASS** (6/6 automated tests passed)

TC-SLOT-02 — Sensor → Occupied + red LED

Field	Detail
Requirement(s)	VPS-F-006 (Slot status Occupied + LED red within 2 s of sensor trigger)
Type / priority	Functional / Performance / High
Preconditions	Slot L1-C001 vacant
Steps	1. POST sensor event occupied=true. 2. POST occupied=false. 3. Replay an old sequence number.
Expected result	1: status OCCUPIED, LED RED within 2000 ms. 2: VACANT, GREEN. 3: 409 SENSOR_REPLAY.
Sprint update	Replay protection added (security objective 3).
Automated by	`unit/test_sensors.py::test_clear_event_sets_green` `unit/test_sensors.py::test_occupied_event_sets_red_led_within_2s`
Latest result	**PASS** (2/2 automated tests passed)

TC-SLOT-03 — VMS boards match database

Field	Detail
Requirement(s)	VPS-F-007 (Refresh VMS vacancy counts within 1 s)
Type / priority	Functional / Medium
Preconditions	Default layout
Steps	1. Occupy 3 bays via sensors. 2. Read /api/v1/vms. 3. Fill every bay.
Expected result	Entrance count equals DB vacant count; version increments per change; boards show 'FULL' when full.
Sprint update	Boards cached per slot-state version (performance).
Automated by	`unit/test_sensors.py::test_vms_counts_match_database` `unit/test_sensors.py::test_vms_full_text`
Latest result	**PASS** (2/2 automated tests passed)

TC-SLOT-04 — Lot full / concurrent allocation

Field	Detail
Requirement(s)	VPS-F-005 (Designate nearest vacant slot by vehicle category)
Type / priority	Negative / Concurrency / High
Preconditions	Default layout
Steps	1. Fill all 4 HANDICAP bays, request a 5th. 2. 90 threads allocate 4W concurrently.
Expected result	1: 409 LOT_FULL. 2: exactly 68 distinct slots allocated, 22 LOT_FULL, no double booking.
Sprint update	New case (Sprint 1).

Field	Detail

Automated by	`system/test_end_to_end.py::test_full_category_rejected_cleanly`
	`unit/test_allocation.py::test_concurrent_allocation_never_double_books`
	`unit/test_allocation.py::test_lot_full`
	`unit/test_ingress.py::test_lot_full`

Latest result	**PASS** (4/4 automated tests passed)

TC-BILL-01 — Tiered fare computation

Field	Detail
Requirement(s)	VPS-F-008 (Fee by duration, category and tiered rate tables)
Type / priority	Functional / High
Preconditions	Default tariffs (4W: Rs 40 first hour, Rs 30/h, cap Rs 400)
Steps	Compute fares for 60, 61, 180 min, 24 h, 26 h (4W) and 2W/EV/HANDICAP samples.
Expected result	Base = first_hour + (h-1) x per_hour, capped per 24 h block; CGST = SGST = 9 %; total = base x 1.18.
Sprint update	Multi-day blocks and GST split clarified.
Automated by	`unit/test_tariff.py::test_exit_before_entry` `unit/test_tariff.py::test_tiered_rates_and_daily_cap`
Latest result	**PASS** (10/10 automated tests passed)

TC-BILL-02 — 15-minute grace period

Field	Detail
Requirement(s)	VPS-F-009 (15-minute grace period at zero charge)
Type / priority	Boundary / Medium
Preconditions	Ticket issued
Steps	Exit at 0, 1, 15 and 16 minutes.
Expected result	0-15 min: Rs 0, no payment needed, barrier opens; 16 min: one hour charged.
Sprint update	Zero-fare exit bypasses the gateway (Sprint 1).
Automated by	`unit/test_egress.py::test_grace_period_exit_needs_no_payment` `unit/test_tariff.py::test_grace_period_boundary`
Latest result	**PASS** (5/5 automated tests passed)

TC-BILL-03 — Surge and coupons

Field	Detail
Requirement(s)	VPS-F-010 (Peak-hour surge rates and promo coupons)
Type / priority	Functional / Medium
Preconditions	Surge rule 17:00-20:00 IST x1.25; coupons WELCOME10 (10 %), FLAT50 (Rs 50)
Steps	1. Exit at 18:00 IST. 2. Apply WELCOME10. 3. Apply FLAT50 to a Rs 20 fare. 4. Expired / exhausted / unknown coupon.
Expected result	1: surge = 25 % of base. 2: 10 % off (base + surge). 3: never below zero. 4: 400 INVALID_COUPON.
Sprint update	Coupon usage limits and expiry added.

Field	Detail

Automated by	`unit/test_tariff.py::TestSurgeAndCoupons::test_coupon_creation_rules`
	`unit/test_tariff.py::TestSurgeAndCoupons::test_flat_coupon_never_negative`
	`unit/test_tariff.py::TestSurgeAndCoupons::test_invalid_expired_exhausted`
	`unit/test_tariff.py::TestSurgeAndCoupons::test_no_surge_outside_window`
	`unit/test_tariff.py::TestSurgeAndCoupons::test_percent_coupon`
	`unit/test_tariff.py::TestSurgeAndCoupons::test_surge_in_peak_window`
	`unit/test_tariff.py::TestSurgeAndCoupons::test_surge_rule_management`

Latest result	**PASS** (7/7 automated tests passed)

TC-PAY-01 — Exit scan retrieves invoice

Field	Detail
Requirement(s)	VPS-F-011 (Exit QR / plate scan retrieves invoice within 1 s)
Type / priority	Functional / Performance / High
Preconditions	Vehicle parked 2 h 10 min
Steps	1. Scan QR at EXIT-1. 2. Look up by plate. 3. Unknown plate.
Expected result	Invoice (3 billable hours, Rs 118.00) in <= 1000 ms; plate lookup finds the same ticket; unknown → 404.
Sprint update	Unchanged.
Automated by	`unit/test_egress.py::test_lookup_by_qr_and_plate`
Latest result	**PASS** (1/1 automated tests passed)

TC-PAY-02 — Cashless payment

Field	Detail
Requirement(s)	VPS-F-012 (Cashless payment (UPI QR, cards, wallets); webhook PAID within 3 s)
Type / priority	Functional / Integration / High
Preconditions	Invoice retrieved
Steps	1. Checkout with UPI → order + dynamic UPI QR. 2. Gateway sends signed webhook payment.captured.
Expected result	Order CREATED with upi:// URI; webhook verified → ticket PAID within 3 s.
Sprint update	Async PENDING → PAID flow (DEV-05).
Automated by	`unit/test_egress.py::test_upi_payment_opens_barrier_and_releases_slot` `unit/test_payment.py::test_upi_order_has_dynamic_qr`
Latest result	**PASS** (2/2 automated tests passed)

TC-PAY-03 — Exit barrier on PAID only

Field	Detail
Requirement(s)	VPS-F-013 (Raise exit barrier only on verified payment)
Type / priority	Functional / Security / High
Preconditions	Vehicle parked, unpaid
Steps	1. Try to open exit barrier. 2. Pay. 3. Observe barrier and slot.
Expected result	1: 403 UNPAID_TICKET, barrier DOWN. 3: barrier UP, slot VACANT, EXIT event queued to central.
Sprint update	Unchanged.
Automated by	`unit/test_egress.py::test_barrier_never_opens_unpaid` `unit/test_egress.py::test_upi_payment_opens_barrier_and_releases_slot`
Latest result	**PASS** (2/2 automated tests passed)

TC-PAY-04 — GST tax receipt

Field	Detail
Requirement(s)	VPS-F-014 (Itemised GST receipt via printer, e-mail or SMS)
Type / priority	Functional / Medium
Preconditions	Ticket PAID
Steps	1. GET receipt JSON. 2. GET PDF. 3. Send by SMS / e-mail; invalid recipients.
Expected result	Invoice no. VPS//, GSTIN, SAC 996743, CGST 9 % + SGST 9 %; valid PDF; SMS queued with masked recipient stored encrypted; invalid channel/recipient → 400.
Sprint update	Moved to Sprint 2.

Automated by	`unit/test_receipt.py::TestReceipts::test_grace_receipt_and_unpaid`
	`unit/test_receipt.py::TestReceipts::test_gst_invoice`
	`unit/test_receipt.py::TestReceipts::test_pdf_and_sms`
	`unit/test_receipt.py::test_receipt_api`

Latest result	**PASS** (4/4 automated tests passed)

TC-PAY-05 — Payment failure & webhook security

Field	Detail
Requirement(s)	VPS-F-012 (Cashless payment (UPI QR, cards, wallets); webhook PAID within 3 s), VPS-F-013 (Raise exit barrier only on verified payment)
Type / priority	Negative / Security / High
Preconditions	Order created
Steps	1. Gateway reports payment.failed. 2. Retry. 3. Re-deliver the same webhook. 4. Tamper amount / bad signature.
Expected result	1: FAILED, barrier stays DOWN, attendant alert. 2: PAID. 3: idempotent. 4: 403.
Sprint update	New case (Sprint 1).

Automated by	`unit/test_egress.py::test_failed_payment_keeps_barrier_closed`
	`unit/test_egress.py::test_webhook_idempotent_and_amount_checked`
	`unit/test_payment.py::test_webhook_signature`

Latest result	**PASS** (3/3 automated tests passed)

TC-RES-01 — 48-hour advance reservation

Field	Detail
Requirement(s)	VPS-F-015 (Reserve a slot up to 48 h in advance)
Type / priority	Functional / Boundary / High
Preconditions	Vacant bay of the category
Steps	1. Book 2 h ahead. 2. Book 47.9 h / 48.1 h / in the past. 3. Second booking for the same plate.
Expected result	1: BOOKED, slot RESERVED (blue LED), 8-char code. 2: 47.9 h OK, others 400 INVALID_WINDOW. 3: 409.
Sprint update	One active booking per vehicle added.

Automated by	`unit/test_reservation.py::TestBooking::test_48_hour_window`
	`unit/test_reservation.py::TestBooking::test_input_validation`
	`unit/test_reservation.py::TestBooking::test_lookup_list_cancel`
	`unit/test_reservation.py::TestBooking::test_one_active_booking_per_vehicle`
	`unit/test_reservation.py::TestBooking::test_reserve_locks_slot_and_issues_code`

Latest result	**PASS** (5/5 automated tests passed)

TC-RES-02 — No-show auto-release

Field	Detail
Requirement(s)	VPS-F-016 (Release reserved slot 30 min after start on no-show)

Field	Detail
Type / priority	Functional / Boundary / Medium
Preconditions	Booking for T+1 h
Steps	1. Run release at start+29 min. 2. Run at start+31 min.
Expected result	1: nothing released. 2: reservation EXPIRED, slot VACANT, audit entry NO_SHOW_RELEASED.
Sprint update	Background scheduler every 60 s.
Automated by	`unit/test_reservation.py::TestNoShow::test_release_after_30_minutes` `unit/test_reservation.py::TestNoShow::test_scheduler_tick_and_thread`
Latest result	**PASS** (2/2 automated tests passed)

TC-RES-03 — Reservation check-in at entry

Field	Detail
Requirement(s)	VPS-F-015 (Reserve a slot up to 48 h in advance)
Type / priority	Functional / Negative / Medium
Preconditions	Booking exists
Steps	1. Enter with code and matching plate. 2. Wrong plate / wrong category. 3. > 60 min early. 4. > 30 min late.
Expected result	1: reserved slot assigned, CHECKED_IN. 2: 403 / 400. 3: 409 RESERVATION_TOO_EARLY. 4: 409 RESERVATION_EXPIRED.
Sprint update	New case (Sprint 2).

Automated by	`unit/test_reservation.py::TestCheckIn::test_entry_with_code_uses_reserved_slot`
	`unit/test_reservation.py::TestCheckIn::test_late_arrival_expires`
	`unit/test_reservation.py::TestCheckIn::test_plate_and_category_must_match`
	`unit/test_reservation.py::TestCheckIn::test_too_early`

Latest result	**PASS** (4/4 automated tests passed)

TC-ADM-01 — Admin configuration without restart

Field	Detail
Requirement(s)	VPS-F-017 (Admin console: tariffs, zones, attendant credentials; no restart)
Type / priority	Functional / High
Preconditions	Admin signed in with TOTP
Steps	1. Change 4W tariff. 2. Look up an active ticket. 3. Add surge rule, coupon, slots; disable a bay; create attendant.
Expected result	New tariff applies immediately; all changes audited; manager is read-only (403 on writes).
Sprint update	Layout and credential management added.

Automated by	`unit/test_admin_api.py::TestLayoutAdmin::test_add_slots_and_status`
	`unit/test_admin_api.py::TestTariffAdmin::test_invalid_tariff`
	`unit/test_admin_api.py::TestTariffAdmin::test_manager_read_only`
	`unit/test_admin_api.py::TestTariffAdmin::test_surge_and_coupons`
	`unit/test_admin_api.py::TestTariffAdmin::test_update_tariff_live`
	`unit/test_tariff.py::test_tariff_update_takes_effect_without_restart`

Latest result	**PASS** (6/6 automated tests passed)

TC-ADM-02 — Report export <= 5 s

Field	Detail
Requirement(s)	VPS-F-018 (Revenue / turnaround / occupancy reports in CSV/PDF within 5 s)
Type / priority	Functional / Performance / Medium
Preconditions	Paid sessions exist (incl. 5,000-row month)

Field	Detail
Steps	Export revenue, occupancy, turnaround as CSV, PDF, JSON (daily/monthly).
Expected result	Correct aggregates; every export <= 5000 ms; attendant gets 403.
Sprint update	5,000-row volume test added.

Automated by	`perf/test_performance.py::test_export_large_month_under_5s`
	`unit/test_reports.py::TestReports::test_bad_dates`
	`unit/test_reports.py::TestReports::test_empty_exports`
	`unit/test_reports.py::TestReports::test_export_under_5_seconds`
	`unit/test_reports.py::TestReports::test_occupancy`
	`unit/test_reports.py::TestReports::test_out_of_range_rows_excluded`
	`unit/test_reports.py::TestReports::test_revenue_daily`
	`unit/test_reports.py::TestReports::test_revenue_monthly_and_range`
	... +2 more

Latest result	**PASS** (20/20 automated tests passed)

TC-Perf-01 — Ingress latency p90

Field	Detail
Requirement(s)	VPS-NF-001 (Ingress transaction p90 <= 3 s)
Type / priority	Performance / High
Preconditions	Server process with 10,000 bays
Steps	200 sequential HTTP ingress transactions; measure round trip.
Expected result	p90 <= 3000 ms.
Sprint update	JMeter replaced by pytest/httpx harness (DEV-06).
Automated by	`perf/test_performance.py::test_ingress_p90_under_3s`
Latest result	**PASS** (1/1 automated tests passed)

TC-Rel-01 — Availability proxy (soak)

Field	Detail
Requirement(s)	VPS-NF-002 (99.9 % monthly availability)
Type / priority	Reliability / High
Preconditions	Fresh system
Steps	600+ mixed entry/exit/display requests; check health.
Expected result	Zero 5xx responses; >= 100 completed exits; /health ok.
Sprint update	Lab proxy for monthly uptime (DEV-10).
Automated by	`system/test_resilience.py::test_soak_no_server_errors`
Latest result	**PASS** (1/1 automated tests passed)

TC-Scale-01 — Scale: 10k slots / 50 lanes

Field	Detail
Requirement(s)	VPS-NF-003 (10,000 slots, 50 lanes, < 200 ms API latency at 500 req/s)
Type / priority	Performance / Load / High
Preconditions	Separate server process, 10,000 bays, 25 lane pairs
Steps	1,000 mixed requests (ingress, sensor, VMS, nearest-slot) from 25 concurrent clients after warm-up.
Expected result	Zero 5xx; p95 < 200 ms; throughput reported (target 500 req/s).
Sprint update	Server run out-of-process.

Field	Detail
Automated by	`perf/test_performance.py::test_api_latency_under_load_10k_slots_50_lanes`
Latest result	**PASS** (1/1 automated tests passed)

TC-SEC-06 — PCI-DSS: no plaintext card data

Field	Detail
Requirement(s)	VPS-NF-004 (PCI-DSS: no plaintext card data or credentials)
Type / priority	Security / High
Preconditions	Card payment completed
Steps	Dump the database after card payment, SMS receipt, reservation with contact.
Expected result	No PAN, plate, phone, password or TOTP secret in plaintext; only card last-4 stored.
Sprint update	Renamed from TC-Sec-01 (DEV-08).
Automated by	`system/test_security_validation.py::test_database_dump_has_no_plaintext_pii_or_pan` `unit/test_payment.py::test_card_pan_never_stored`
Latest result	**PASS** (2/2 automated tests passed)

TC-Fault-01 — Edge offline resilience

Field	Detail
Requirement(s)	VPS-NF-005 (72 h autonomous edge queuing; auto-sync <= 5 min)
Type / priority	Resilience / High
Preconditions	WAN link online
Steps	1. Cut WAN. 2. Run 30 entries + 20 exits over ~72 h. 3. Restore WAN. 4. Lane controller offline. 5. Gateway outage.
Expected result	2: lanes keep working, 50 events queued, within 72 h budget. 3: all synced in order <= 5 min, no duplicates. 4: 503 EDGE_CONTROLLER_OFFLINE on that lane only. 5: 402 → cash fallback works.
Sprint update	Gateway outage scenario added.

Automated by	`system/test_resilience.py::test_lane_controller_offline_returns_503_and_other_lanes_continue`
	`system/test_resilience.py::test_lanes_keep_working_through_72h_wan_outage`
	`system/test_resilience.py::test_payment_gateway_outage_falls_back_to_cash`
	`unit/test_edge.py::TestEdgeQueue::test_offline_budget_72h`
	`unit/test_edge.py::TestEdgeQueue::test_offline_queues_then_syncs_in_order`
	`unit/test_edge.py::TestEdgeQueue::test_online_forwards_immediately`
	`unit/test_edge.py::TestEdgeQueue::test_replay_is_idempotent`
	`unit/test_edge_api.py::test_edge_api`

Latest result	**PASS** (8/8 automated tests passed)

TC-SEC-01 — TLS 1.3 enforced

Field	Detail
Requirement(s)	VPS-SR-001 (TLS 1.3+ mandatory on all connections)
Type / priority	Security / High
Preconditions	Server started with --tls
Steps	1. HTTPS with TLS 1.3. 2. Client limited to TLS 1.2. 3. Plain HTTP.
Expected result	1: 200, negotiated TLSv1.3, HSTS header. 2: handshake refused. 3: not served (426 behind proxy).
Sprint update	Real handshake test against a live server.

Field	Detail

Automated by	`system/test_security_validation.py::TestTLS::test_older_tls_rejected`
	`system/test_security_validation.py::TestTLS::test_plain_http_rejected`
	`system/test_security_validation.py::TestTLS::test_tls13_handshake_succeeds`
	`unit/test_app.py::test_plain_http_refused_when_tls_required`
	`unit/test_app.py::test_security_headers`
	`unit/test_tls.py::test_config_pins_tls13`
	`unit/test_tls.py::test_self_signed_cert_created_once`

Latest result	**PASS** (7/7 automated tests passed)

TC-SEC-02 — Password hashing & PII encryption

Field	Detail
Requirement(s)	VPS-SR-002 (Argon2id/bcrypt passwords; AES-256 PII at rest)
Type / priority	Security / High
Preconditions	Users seeded
Steps	Inspect stored password hashes, TOTP secrets, plates, contacts.
Expected result	Passwords $argon2id$; PII AES-256-GCM (random nonce); wrong key fails to decrypt.
Sprint update	Unchanged.

Automated by	`system/test_security_validation.py::test_database_dump_has_no_plaintext_pii_or_pan`
	`unit/test_auth.py::test_secrets_not_stored_in_plaintext`
	`unit/test_security.py::TestPIIVault::test_encrypt_decrypt`
	`unit/test_security.py::TestPIIVault::test_fingerprint_deterministic_and_keyed`
	`unit/test_security.py::TestPIIVault::test_key_must_be_256_bit`
	`unit/test_security.py::TestPIIVault::test_nonce_is_random`
	`unit/test_security.py::TestPIIVault::test_unknown_format`
	`unit/test_security.py::TestPIIVault::test_wrong_key_cannot_decrypt`
	… +3 more

Latest result	**PASS** (11/11 automated tests passed)

TC-SEC-03 — RBAC + MFA

Field	Detail
Requirement(s)	VPS-SR-003 (RBAC + MFA for administrative accounts)
Type / priority	Security / High
Preconditions	admin/manager/attendant accounts
Steps	1. Admin login without / with wrong / with valid TOTP. 2. RBAC matrix over protected endpoints. 3. 5 bad passwords.
Expected result	1: MFA_REQUIRED / 401 / 200. 2: 401 anonymous, 403 wrong role. 3: ACCOUNT_LOCKED.
Sprint update	Lockout and session expiry added.

Automated by	`system/test_security_validation.py::test_mfa_enforced_over_api`
	`system/test_security_validation.py::test_rbac_matrix`
	`unit/test_app.py::test_rbac_guards`
	`unit/test_auth.py::test_admin_login_with_totp`
	`unit/test_auth.py::test_admin_requires_totp`
	`unit/test_auth.py::test_attendant_password_only`
	`unit/test_auth.py::test_authorize`
	`unit/test_auth.py::test_create_user_rules`
	… +14 more

Latest result	**PASS** (29/29 automated tests passed)

TC-SEC-04 — Counterfeit QR rejected

Field	Detail
Requirement(s)	VPS-SR-004 (HMAC-SHA256 signed QR tickets; tampered tickets rejected)
Type / priority	Security / High
Preconditions	Ticket issued
Steps	1. Modify one character of the QR body. 2. Re-sign a different slot with a guessed key.
Expected result	403 TAMPERED_TICKET; CRITICAL alert to attendant; audit entry COUNTERFEIT_TICKET.
Sprint update	Unchanged.

Automated by	`system/test_security_validation.py::test_counterfeit_qr_rejected_with_alert`
	`unit/test_egress.py::test_counterfeit_ticket_rejected_and_alerted`
	`unit/test_security.py::TestTicketSigner::test_malformed_payload`
	`unit/test_security.py::TestTicketSigner::test_other_key_rejected`
	`unit/test_security.py::TestTicketSigner::test_roundtrip`
	`unit/test_security.py::TestTicketSigner::test_signature_is_hmac_sha256_hex`
	`unit/test_security.py::TestTicketSigner::test_tampered_body_rejected`
	`unit/test_security.py::TestTicketSigner::test_tampered_signature_rejected`

Latest result	**PASS** (10/10 automated tests passed)

TC-SEC-05 — Immutable audit log

Field	Detail
Requirement(s)	VPS-SR-005 (Immutable write-once audit log retained 3 years)
Type / priority	Security / High
Preconditions	Audited actions performed
Steps	1. Verify chain. 2. UPDATE / DELETE an entry via SQL. 3. Drop the trigger and edit a row.
Expected result	1: valid. 2: refused (write-once). 3: chain verification reports the broken entry. Retention >= 3 years.
Sprint update	Hash chain added on top of triggers.

Automated by	`system/test_security_validation.py::test_audit_log_tamper_evident`
	`unit/test_admin_api.py::TestOverrideAndAudit::test_audit_verify_endpoint`
	`unit/test_admin_api.py::TestOverrideAndAudit::test_override_is_audited`
	`unit/test_admin_api.py::TestOverrideAndAudit::test_override_validation`
	`unit/test_audit.py::TestAuditLog::test_entries_filter`
	`unit/test_audit.py::TestAuditLog::test_hash_chain`
	`unit/test_audit.py::TestAuditLog::test_out_of_band_tampering_detected`
	`unit/test_audit.py::TestAuditLog::test_retention_three_years`
	… +1 more

Latest result	**PASS** (9/9 automated tests passed)

TC-SYS-01 — Walk-in car, UPI payment

Field	Detail
Requirement(s)	VPS-F-001 (Capture vehicle plate via ANPR within 1.5 s (>= 95 % OCR accuracy)), VPS-F-005 (Designate nearest vacant slot by vehicle category), VPS-F-008 (Fee by duration, category and tiered rate tables), VPS-F-012 (Cashless payment (UPI QR, cards, wallets); webhook PAID within 3 s), VPS-F-013 (Raise exit barrier only on verified payment)
Type / priority	System (E2E) / High
Preconditions	Empty car park
Steps	Enter → loop → park (sensor) → 2 h 10 min → scan → UPI → webhook → receipt.
Expected result	VMS decrements then restores; Rs 118.00 charged; barrier opens; receipt total matches.
Sprint update	New.
Automated by	`system/test_end_to_end.py::test_walk_in_car_pays_by_upi`
Latest result	**PASS** (1/1 automated tests passed)

TC-SYS-02 — Reserved EV, short visit

Field	Detail
Requirement(s)	VPS-F-015 (Reserve a slot up to 48 h in advance), VPS-F-009 (15-minute grace period at zero charge)
Type / priority	System (E2E) / Medium
Preconditions	EV bay free
Steps	Book 20 min ahead → arrive at +25 min with code → leave after 14 min.
Expected result	Reserved bay used; exit free (grace).
Sprint update	New.
Automated by	`system/test_end_to_end.py::test_reserved_ev_short_visit_in_grace`
Latest result	**PASS** (1/1 automated tests passed)

TC-SYS-03 — Peak-hour card payment with coupon

Field	Detail
Requirement(s)	VPS-F-010 (Peak-hour surge rates and promo coupons), VPS-F-012 (Cashless payment (UPI QR, cards, wallets); webhook PAID within 3 s)
Type / priority	System (E2E) / Medium
Preconditions	Surge rule active
Steps	Enter 16:00 IST → exit 18:30 IST → card + FLAT50.
Expected result	Base 100 + surge 25 − 50 = 75 taxable; total Rs 88.50.
Sprint update	New.
Automated by	`system/test_end_to_end.py::test_peak_hour_card_payment_with_coupon`
Latest result	**PASS** (1/1 automated tests passed)

TC-SYS-04 — Unreadable plate, cash exit

Field	Detail
Requirement(s)	VPS-F-001 (Capture vehicle plate via ANPR within 1.5 s (>= 95 % OCR accuracy)), VPS-F-012 (Cashless payment (UPI QR, cards, wallets); webhook PAID within 3 s)
Type / priority	System (E2E) / Medium
Preconditions	Attendant signed in
Steps	Low-confidence read → attendant manual ticket → 50 min → cash.
Expected result	422 then manual ticket; Rs 47.20 collected; audited.
Sprint update	New.
Automated by	`system/test_end_to_end.py::test_unreadable_plate_manual_entry_and_cash`
Latest result	**PASS** (1/1 automated tests passed)

TC-SYS-05 — Admin day

Field	Detail
Requirement(s)	VPS-F-017 (Admin console: tariffs, zones, attendant credentials; no restart), VPS-F-018 (Revenue / turnaround / occupancy reports in CSV/PDF within 5 s)
Type / priority	System (E2E) / Medium
Preconditions	Admin signed in
Steps	Change 2W tariff mid-stay → exit → revenue report + CSV → audit verify.

Field	Detail
Expected result	New tariff billed; report total Rs 29.50; audit chain valid.
Sprint update	New.
Automated by	`system/test_end_to_end.py::test_admin_day`
Latest result	**PASS** (1/1 automated tests passed)